

5.4 Anti-virus Software

5.4.1 Introduction

All Anti-virus (AV) applications operate on similar principles. All have a virus database that contains a list of file patterns that can be used to identify a virus. A pattern refers to a sequence of bits that is unique to a particular file. The second component of an AV is a scan engine which scans files and compares it with the pattern file to verify if there is an infection. While the scan engine can be run when the user explicitly activates it, it can also be set to run automatically any time a file is opened or a program is executed. This latter mode of operation is called a real-time scan.

With an increase in threats from online data transfers, all AV modules have begun incorporating additional capabilities to the basic program that only scans local files. It is rare to find an AV program that does not include capabilities like e-mail scanning (inbound and outbound), downloaded file scanning, spyware scanning, and such.

5.4.2. Avast!

The Avast! Home Edition anti-virus is free for non-commercial use, given that the user has acquired a freely-available registration key from the site within 60 days of installation to continue using the product. Besides the basic virus scanner to protect local folders, Avast! also includes modules to scan online file transfers. There are modules that scan files transferred during regular browsing, peer-to-peer activities, Instant Messaging, communicating through newsgroups, and e-mail. Avast! creates a database of the files in the system when the system is idle. This “Virus Recovery Database” can be used to recover files damaged by a virus.

The free Home Edition only allows using the simple interface



The Avast! Simple User Interface